The environment evolves in discrete time steps according to some dynamical rules, and at each time step the agent takes an action that may influence the next state. After taking each action, the agent receives feedback in the form of a reward signal. We described a mathematical formalization of this called the *Markov decision process* (MDP).

An MDP is a mathematical structure that includes a set of states, *S*, that the environment can be in, and a set of actions, *A*, that the agent can take, which may depend on the particular state of the environment. There is a *reward function*, $R(s_t,a_t,s_{t+1})$, that produces the *reward signal*, given the transition from the current state to the next state and the agent's action. The environment may evolve deterministically or stochastically, but in either case, the agent initially does not know the dynamical rules of the environment, so all state transitions must be described probabilistically from the perspective of the agent.

We therefore have a conditional probability distribution over next states, s_{t+1}, given the current state and the action taken by the agent at the current time step, denoted $\Pr(S_{t+1}|s_t,a_t)$. The agent follows some policy, π, which is a function that maps a probability distribution over actions given the current state s_t: $\pi{:}S \rightarrow \Pr(A)$. The objective of the agent is to take actions that maximize the time-discounted cumulative rewards over some time horizon. The time-discounted cumulative reward is termed the *return* (often denoted with the character *G* or *R*), and is equal to

$$G_t = \sum_t \gamma^t r_t$$

The return, G_t at time *t*, is equal to the sum of discounted rewards for each time step until the end of the episode (for episodic environments) or until the sequence converges for non-episodic environments. The γ factor is a parameter in the interval (0,1) and is the discount rate that determines how quickly the sequence will converge, and thus how much the future is discounted. A discount rate close to 1 will mean that future rewards are given similar weight to immediate rewards (optimizing over the long term),